



Audencia Business School

Database Design for Glamping Saccargia (Sardinia)

B : Departure point for local tours
Group 2

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Agenda

Part 1. Glamping Saccargia – Conceptual Design

-Background and challenge

-Solution

Part 2. Conceptual design

Part 3. Value summary and future outlook

Background

1.

**Glamping Saccargia is located in Codrongianos, Sardinia, near the famous Basilica di Saccargia
Attracts thousands of tourists each year, but most only visit the Basilica.**

2.

**The project aims to connect cultural, historical, and natural heritage with sustainable tourism.
Group B's role: position the glamping as a departure point for local tours to discover the wider area.**





Challenges

1.

Many tourists stop only at the Basilica di Saccargia and don't explore further.

Low visibility of Codrongianos' cultural sites and local services.

2.

Weak connection between valley activities and the village.

Lack of clear guidance for tourists (where to go next).

Limited transport and tour coordination.

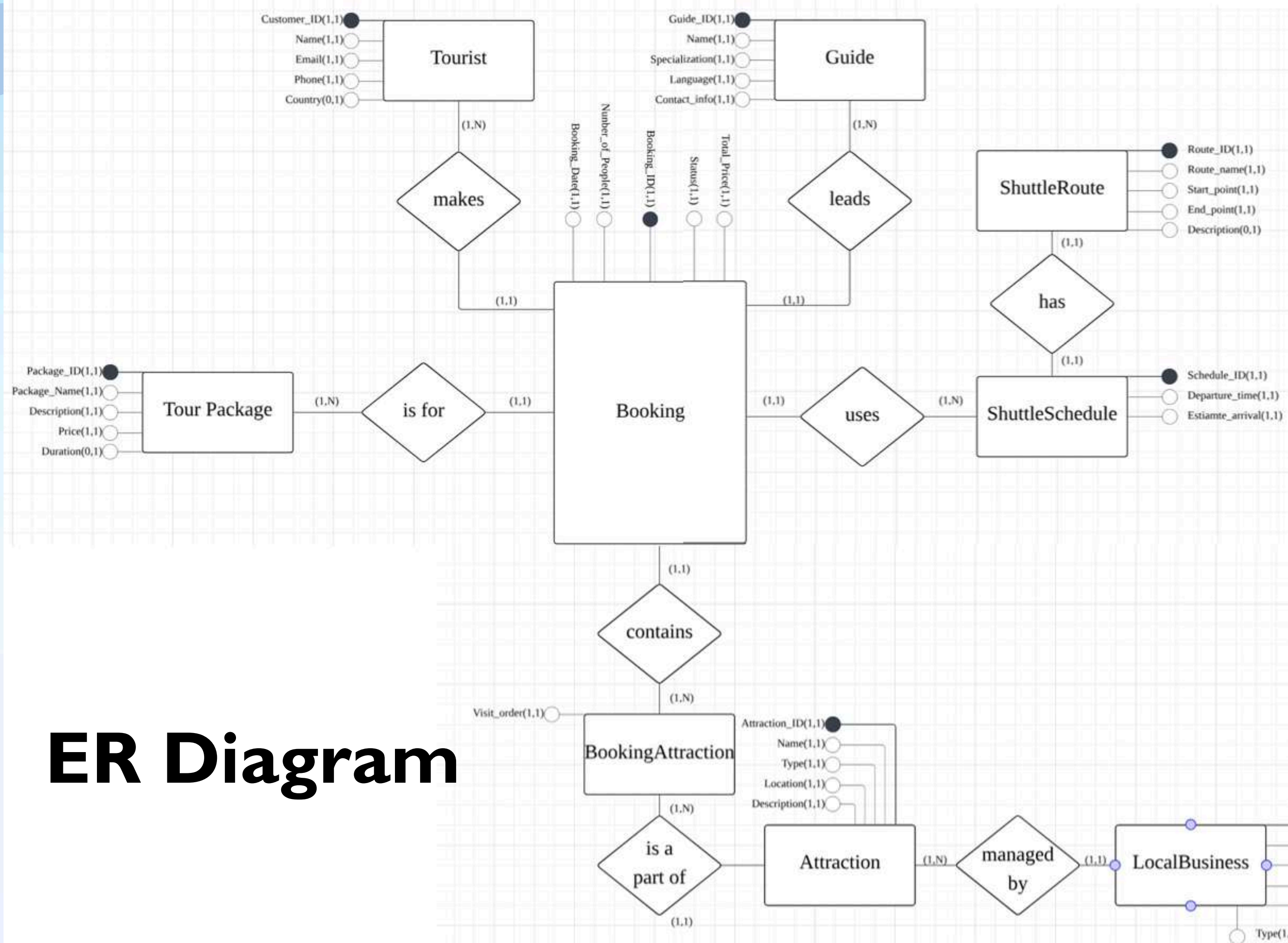
Solution: ER Diagram

Main entities
(Tourists,
Attraction, Local
Business, Booking,
Guide, Shuttle
Schedule)

Connections
between
them

A clear base for
the future
database of
Glamping
Saccargia

ER Diagram

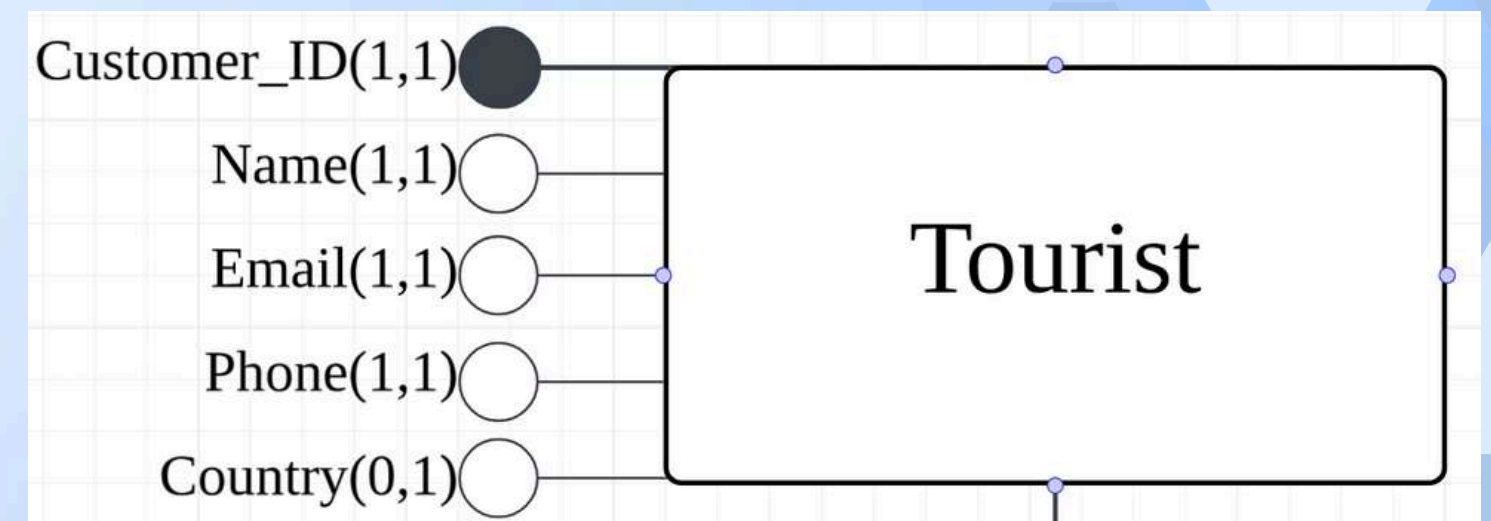


Entity: Tourist

Name Attribute	Attribute Classification*	Type	Dimens	Values and Constraints	Card. MIN	Card. MAX	Candidate?	PK/FK?
Customer_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	PK
Name	Simple	String	20	Not Null	1	1	No	
Email	Simple	String	30	Optional but must be unique (if provided)	1	1	Yes (if Not Null)	
Phone	Simple	String	15	Optional but must be unique (if provided)	1	1	No	
Contry	Simple	String	50	ISO contrylist	0	1	No	

Store main information about each guest

- Each tourist has a unique ID → Customer_ID (Primary Key)
- Attributes:
 - Name → Not Null
 - Email → Unique and not Null
 - Phone → Unique and not Nul
 - Contry → ISO country list
- Email and Phone are optional but must be unique
- Entity will be linked later to bookings or local tour packages

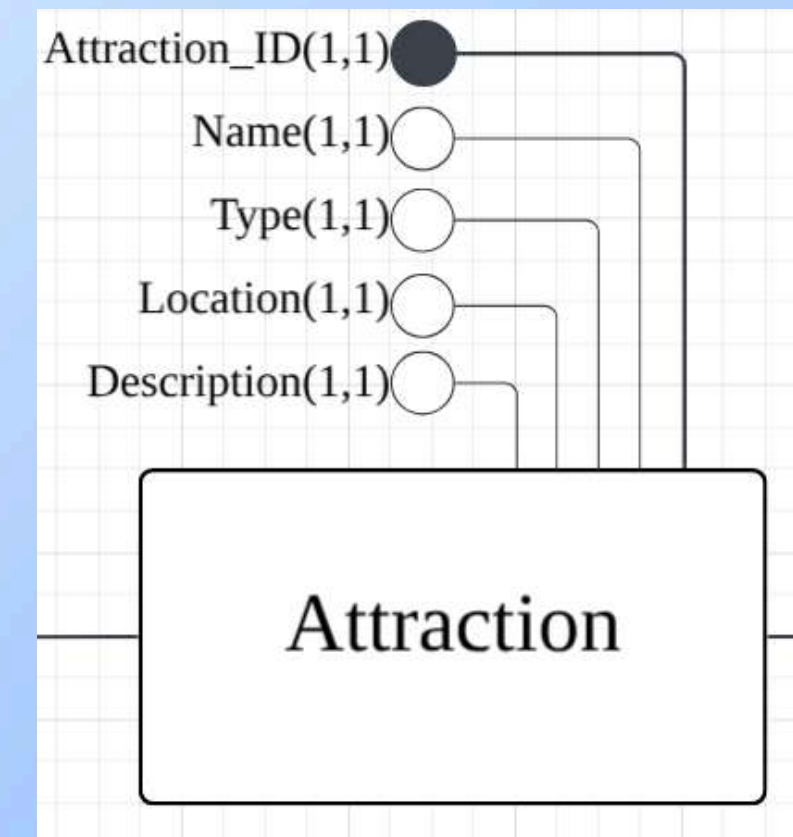


Entity: Attraction

Name Attribute	Attribute Classification*	Type	Dimens	Values and Constraints	Card. MIN	Card. MAX	Candidate?	PK/FK?
Attraction_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	PK
Name	Simple	String	20	Not Null	1	1	No	
Type	Simple	String	20	Enum: church, museum, nature	1	1	No	
Location	Simple	String	20	Adress or coordinates	1	1	No	
Description	Simple	String	50	Optional text field	1	1	No	
Business_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	FK

Represent the main places that tourists can visit

- Each attraction has a unique ID → Attraction_ID (Primary Key)
- Attributes :
 - Name → required (Not Null)
 - Type → enumeration (e.g., church, museum, nature spot...)
 - Location → address or coordinates
 - Description → optional text field
 - Business_ID → Foreign Key linked to the Local Business entity
- Each attraction can be managed by a local business
- The Type attribute helps categorize attractions by their nature

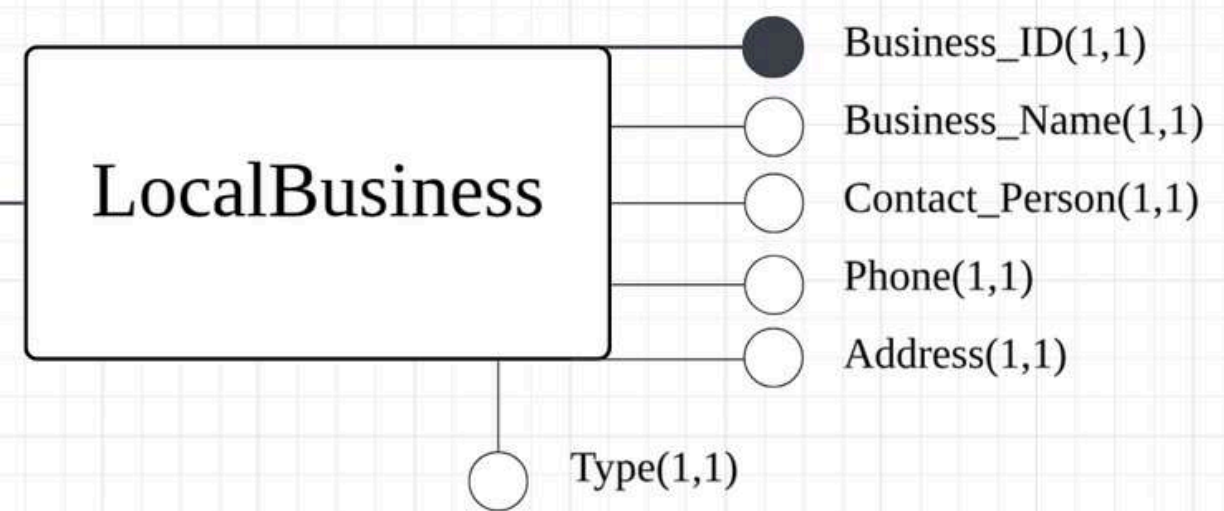


Entity: LocalBusiness

Name Attribute	Attribute Classification*	Type	Dimens	Values and Constraints	Card. MIN	Card. MAX	Candidate?	PK/FK?
Business_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	PK
Business_Name	Simple	String	20	Not Null	1	1	No	
Type	Simple	String	20	Enum: winery museum,	1	1	No	
Contact_Person	Simple	String	20	Optimal contact name	1	1	No	
Phone	Simple	String	15	Unique, optional	1	1	No	
Address	Simple	String	30	Optional business address	1	1	No	

Represent the local partners collaborating with the glamping site

- Each business has a unique ID → Business_ID (Primary Key)
- Attributes :
 - Business Name → required (Not Null)
 - Type → enumeration (e.g., winery, museum, artisan shop, transport...)
 - Contact Person → optional field
 - Phone → unique
 - Address → optional
- This entity is linked to “Attraction” through the Foreign Key Business_ID
- It identifies which local business manages each attraction



Entity: Booking

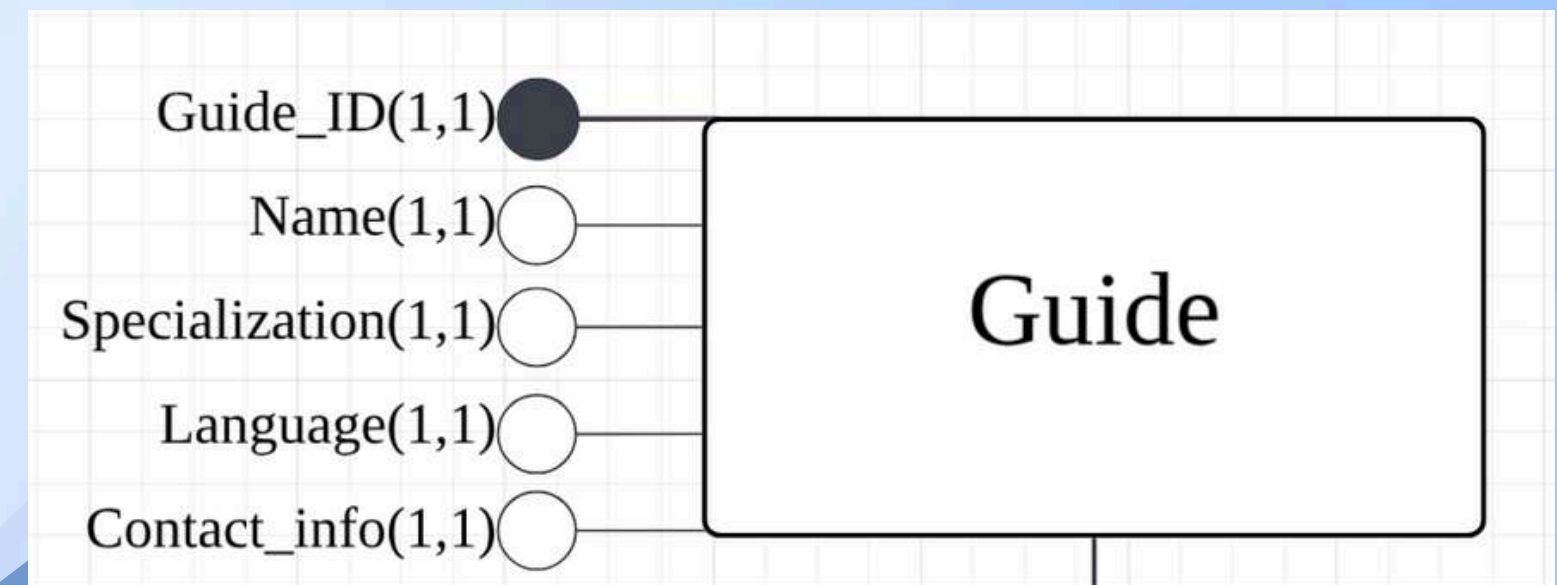
Name Attribute	Attribute Simp./Comp.	Type	Dimens	Values and Constraints	Card. MIN	Card. MAX	Candidate?	PK/FK?
Booking_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	PK
Booking_date	Simple	Date	YYYY-MM-DD	Not Null	1	1	No	-
Number_of_people	Simple	Integer	-	Not Null, >=1	1	1	No	-
Total_price	Simple	Decimal	(10, 2)	Not Null, >=0.00	1	1	No	-
Status	Simple	String	30	Not Null, ['Confirmed', 'Pending', 'Cancelled']	1	1	No	-
Customer_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	FK
Package_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	FK
Guide_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	FK
Schedule_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	FK

- **Central Transaction Record:** Captures all essential details of a customer's reservation, including date, number of people, total price, and booking status.
- **Relational Hub:** Connects multiple entities via foreign keys — Customer_ID, Package_ID, Guide_ID, and Schedule_ID.
- **Status Tracking:** Includes a Status field to monitor booking progress (e.g., Confirmed, Pending, Cancelled)

Entity: Guide

Name Attribute	Attribute Classification	Type	Dimens	Values and Constraints	Card. MIN	Card. MAX	Candidate?	PK/FK?
Guide_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	PK
Name	Simple	String	30	Not Null	1	1	No	-
Specialization	Simple	String	50	Not Null	1	1	No	-
Language	Simple	String	100	Not Null, Comma-separated	1	1	No	-
Contact_info	Simple	String	20	Not Null, Phone or Email	1	1	No	-

- **Personnel Profile:** Stores key information about tour guides, including name, specialization, languages spoken, and contact info.
- **Assigned to Bookings:** Linked to the Booking entity to indicate which guide is responsible for a specific tour.
- **Multilingual Capability:** The Languages attribute supports matching guides with customers based on language preferences.

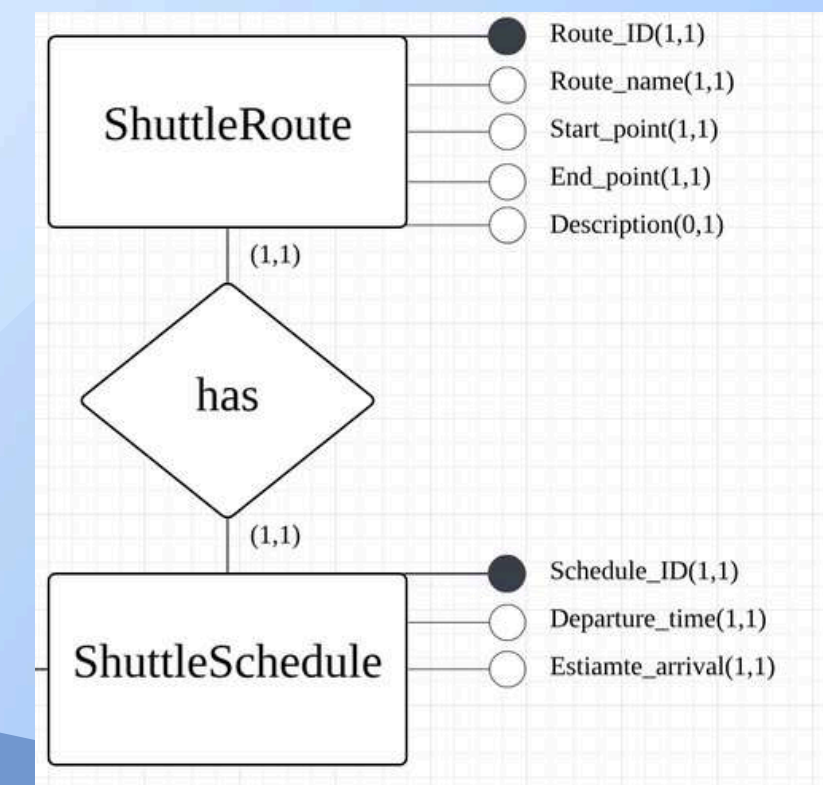


Entity: Shuttle

Shuttle Schedule

Name Attribute	Attribute Classification	Type	Dimens	Values and Constraints	Card. MIN	Card. MAX	Candidate?	PK/FK?
Schedule_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	PK
Departure_time	Simple	Time	HH:MM:SS	Not Null	1	1	No	-
Estimated_arrival	Simple	Time	HH:MM:SS	Not Null	1	1	No	-
Route_id	Simple	String	Auto_Increment	Not Null	1	1	Yes	FK

- **Timing Blueprint:** Defines the departure and estimated arrival times for a specific shuttle trip.
- **Route Linkage:** Connected to a Route_ID to specify the path taken.
- **Referenced in Bookings:** Used in the Booking entity to assign a specific travel schedule to a reservation.



Entity: Shuttle Route

Name Attribute	Attribute Classification	Type	Dimens	Values and Constraints	Card. MIN	Card. MAX	Candidate?	PK/FK?
Route_ID	Simple	String	Auto_Increment	Not Null	1	1	Yes	PK
Route_name	Simple	String	30	Not Null	1	1	No	-
Description	Simple	String	50	Not Null	1	1	No	-
Start_point	Simple	String	50	Not Null	1	1	No	-
End_point	Simple	String	50	Not Null	1	1	No	-

- **Path Definition:** Describes the travel route with a name, description, start point, and end point.
- **Supports Scheduling:** Each route is referenced by one or more schedules to define when and how often it runs.
- **Geographic Clarity:** Helps customers and guides understand the journey's scope and direction.

Conclusion & Future Outlook

1.

→ **Value Created**

- Better tourist guidance beyond the Basilica
- Stronger links with local culture & economy
- Centralized system for efficient management

2.

→ **Future Outlook**

- Expansion of tours and partnerships
- Use of the database for personalized experiences
- Support Codrongianos' sustainable development

The background features a soft rainbow gradient transitioning from light blue on the left to yellow in the center and back to blue on the right. Overlaid on this are various geometric shapes: large, dark blue angular shapes at the top and bottom, and clusters of smaller white and light blue hexagons on the left and right sides.

Thank You